



Faculty of Engineering
Computer Department

Cairo University



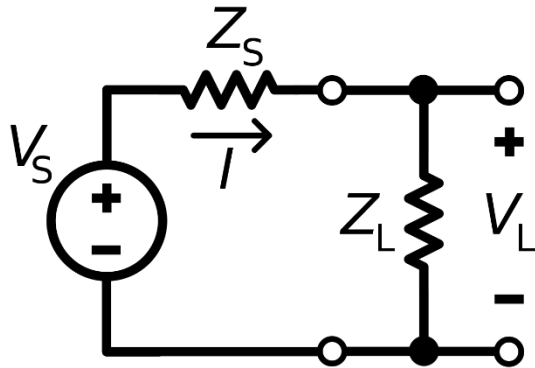
ELC3060

Analog Integrated Circuits

Dr. Amr Hafez

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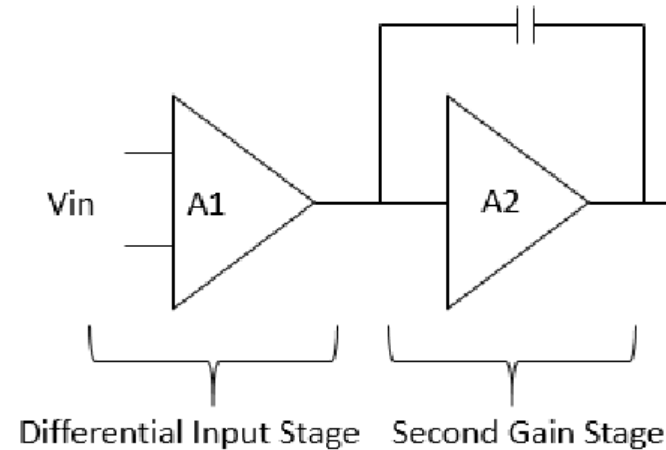
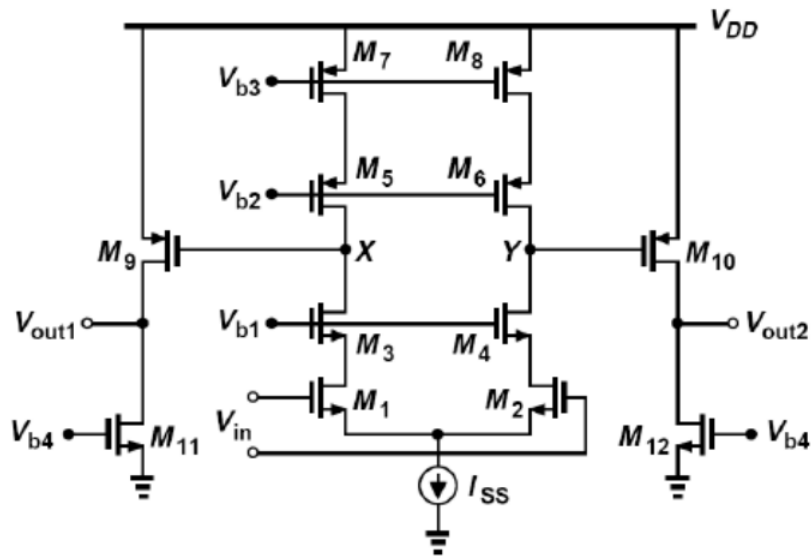
Output Resistance of a Circuit



$$V_L = V_S \frac{Z_L}{Z_L + Z_S}$$

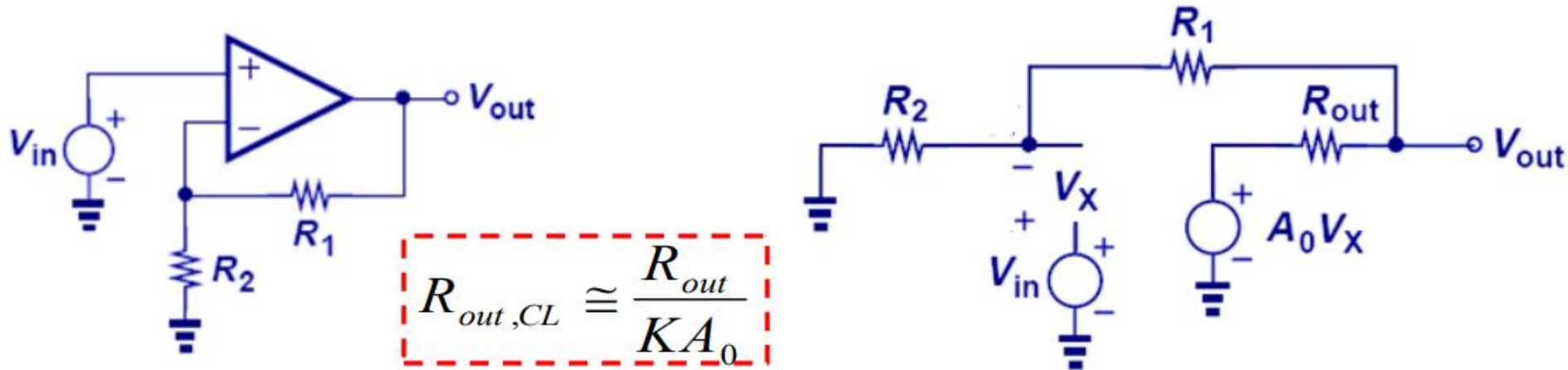
- Z_S should be small with respect to Z_L , otherwise:
 - Voltage loss
 - Power loss
- Z_S can also represent the output impedance of an amplifier which is driving a load, Z_L .
- Achieving a low output resistance is necessary when you are driving heavy loads (when Z_L is small). Example: Audio speakers, or driving large power into an antenna.

The Operational Transconductance Amplifier. OTA



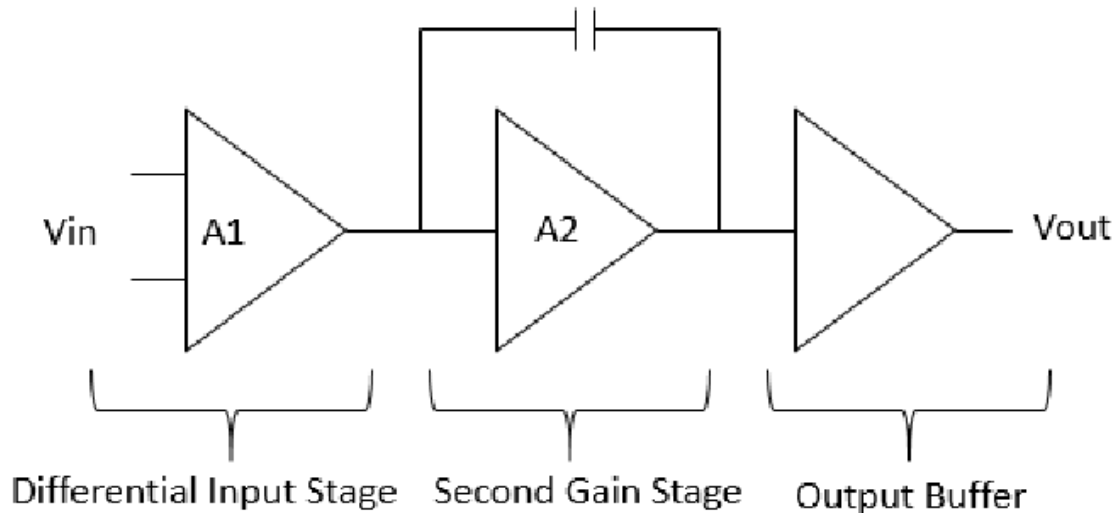
- Both the single-stage and 2-stage amplifiers have a high output resistance (as this is what gives it high gain).
- An OTA is a single-stage or 2-stage differential amplifier that can be used as an OpAmp, but it has a high output resistance.
- But is an OTA any good, if it has a high (sometimes very high) output resistance? Answer is yes, if it drives capacitive loads or large load resistances (those will not affect the DC gain of the OTA. Also, the feedback reduces the output resistance).

Output Resistance and Feedback



- Prove that the output resistance of the overall circuit is reduced by the loop gain ($K A_0$).
- Feedback reduces output resistance → In many applications, this is sufficient to drive the load.
- But sometimes the load is a small resistance (heavy load) that will kill the gain of the OTA.
- And sometimes, the resulting losses in the OTA are too large to tolerate.
- In such cases, a proper output stage must be added in order to have a proper Op Amp capable of driving heavy loads.

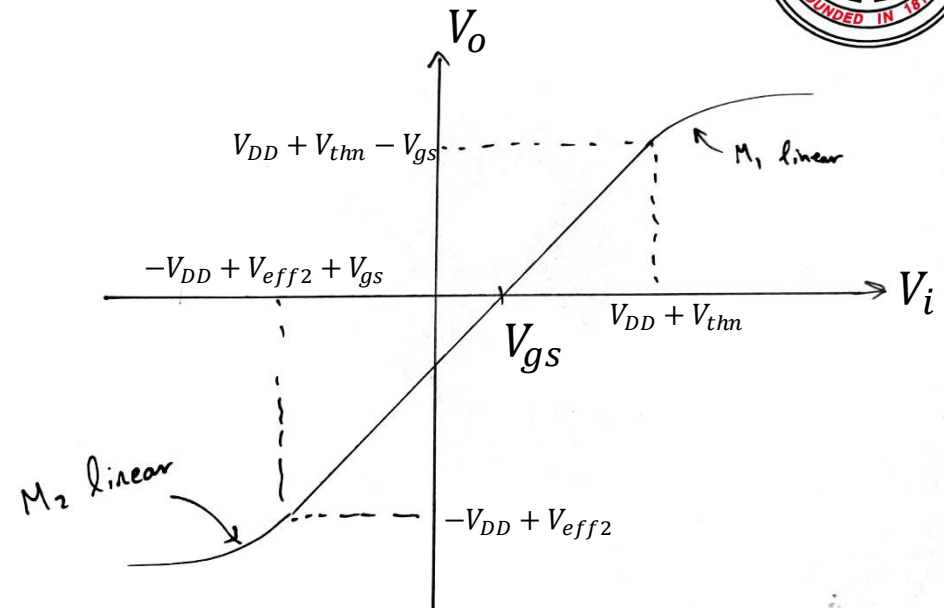
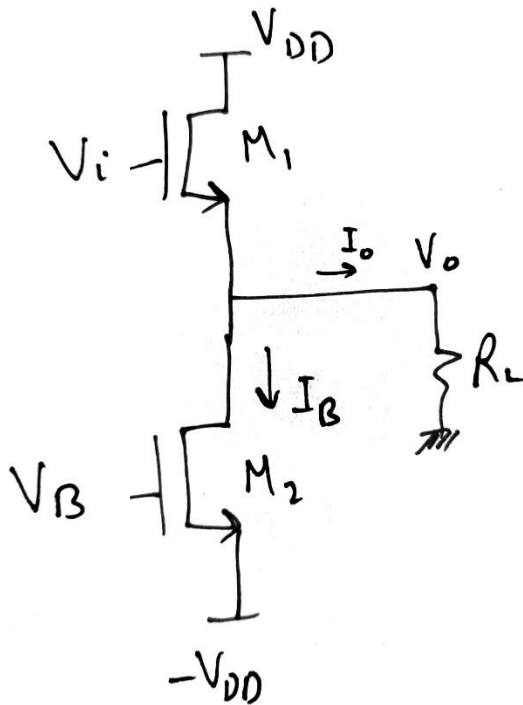
The Operational Amplifier, Op Amp



- The output buffer (also referred to as 'Output Stage') usually gives a voltage gain of unity, but provides a low output resistance (provides power gain but not voltage gain).
- Most of the power consumption of the OpAmp will occur in the output stage
- The main specification on the output stage is:
 - Linearity: does the output stage distort the signal or does it reproduce it faithfully?
 - Efficiency, defined as power delivered to the load divided by power drawn from the supply

$$\eta = \frac{P_L}{P_i}$$

Class A Output Stage: the Common Drain Amplifier



Small signal gain: $v_o = v_i \frac{g_m(R_L \parallel r_o)}{1 + g_m(R_L \parallel r_o)} \approx 1$

$$R_{out} \approx 1/g_m$$

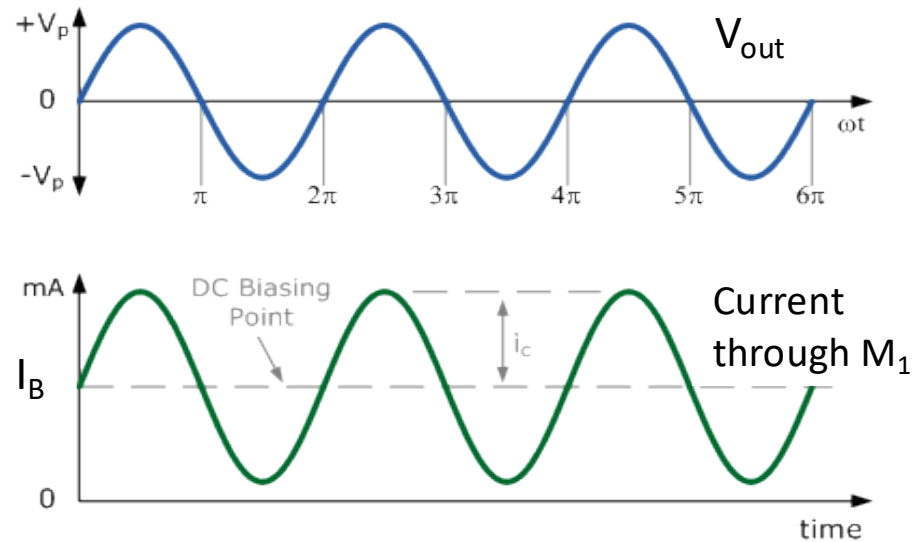
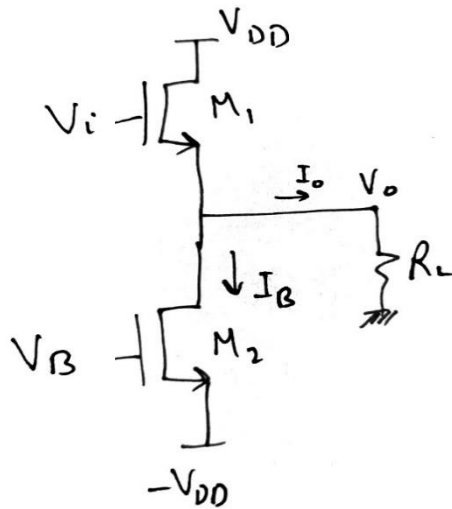
Large signal gain: $V_o = V_i - V_{gs}$; $V_{gs} = V_{th} + \sqrt{\frac{2V_o}{\mu_n C_{ox} \left(\frac{W}{L}\right) R_L}}$

The common drain amplifier provides $\sim$ unity voltage gain and low output resistance. Can provide large current to load, which comes from the supplies.

Class A Output Stage: Efficiency

Suppose V_i , V_o are a sine wave $\rightarrow V_o = \hat{V} \cos \omega t$ (assume no d.c. component)

and therefore, $I_o = \frac{\hat{V}}{R_L} \cos \omega t$ (positive and negative) $\rightarrow I_B$ *must be* $\geq \frac{\hat{V}}{R_L}$



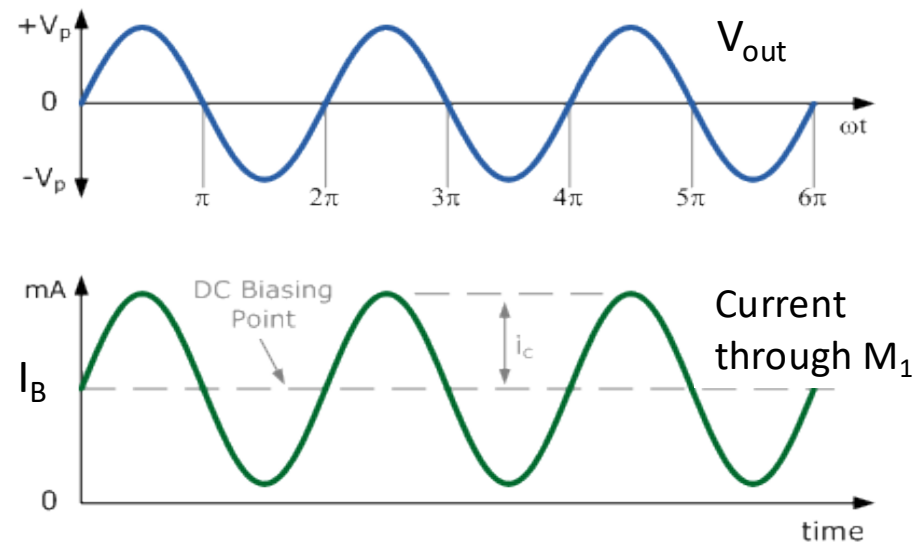
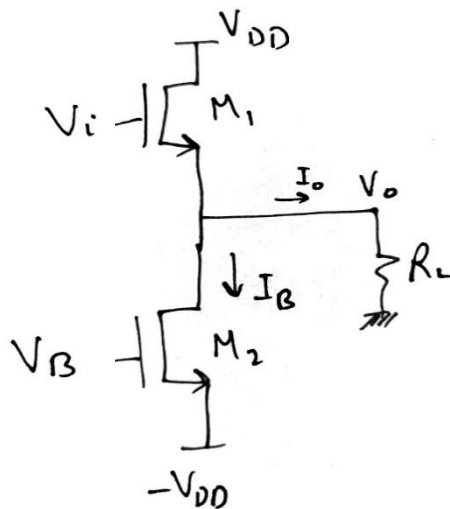
Power delivered to load: $P_L = \frac{\hat{V}^2}{2R_L}$ W

Power consumption: $P_S = 2V_{DD}I_B \geq 2V_{DD}\hat{V}/R_L$ W

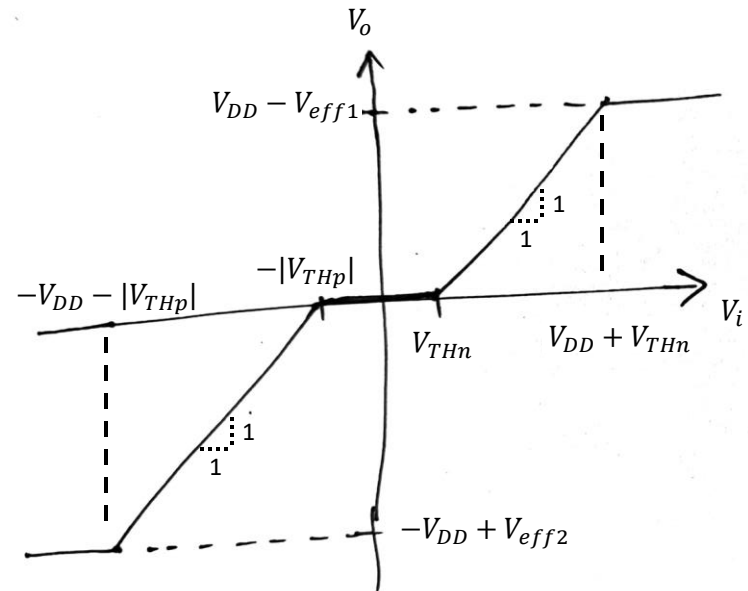
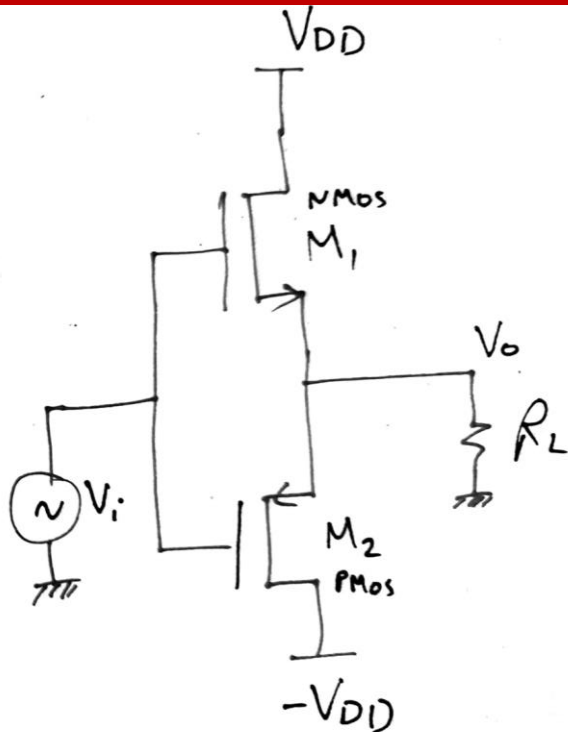
Efficiency: $\eta = P_L/P_S = \hat{V}/(4V_{DD}) < 25\% \rightarrow$ **poor**

Class A Output Stage

- Class A output stage exhibits good linearity, but it has a poor efficiency.
- Looking closely, we see that both M_1 and M_2 are ON all of the time (conduction angle $\theta = 360^\circ$)
- But does M_2 need to be ON when output is > 0 ? Does M_1 need to be ON when output is < 0 ?



Class B Output Stage: Push Pull

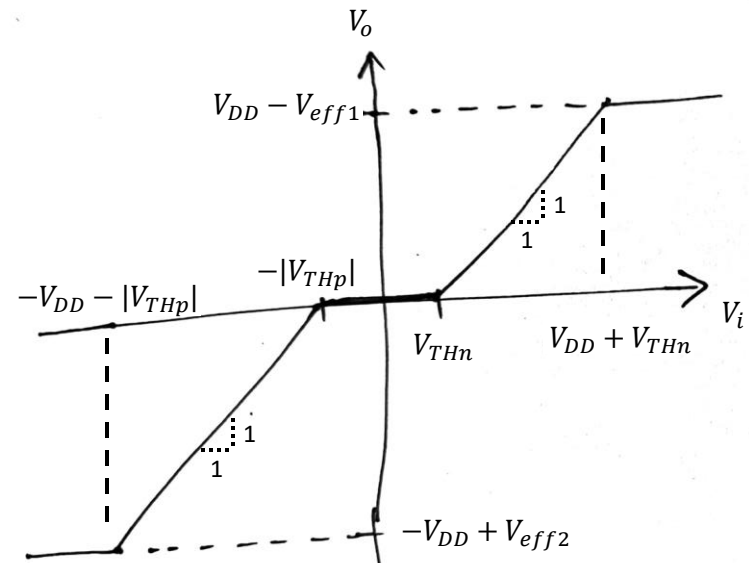
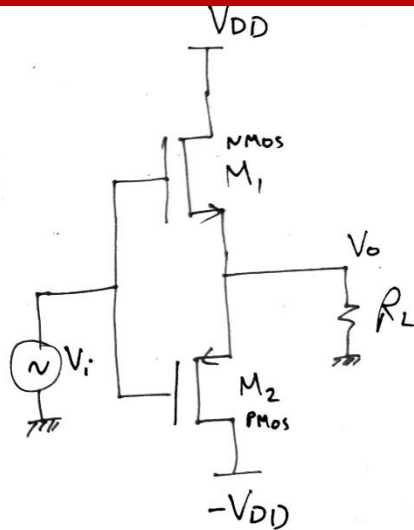


Class B output stage is a push-pull architecture: assume V_i is a sine wave with no d.c. component $\rightarrow$ When V_i is positive, NMOS is ON but PMOS is OFF, and vice versa.

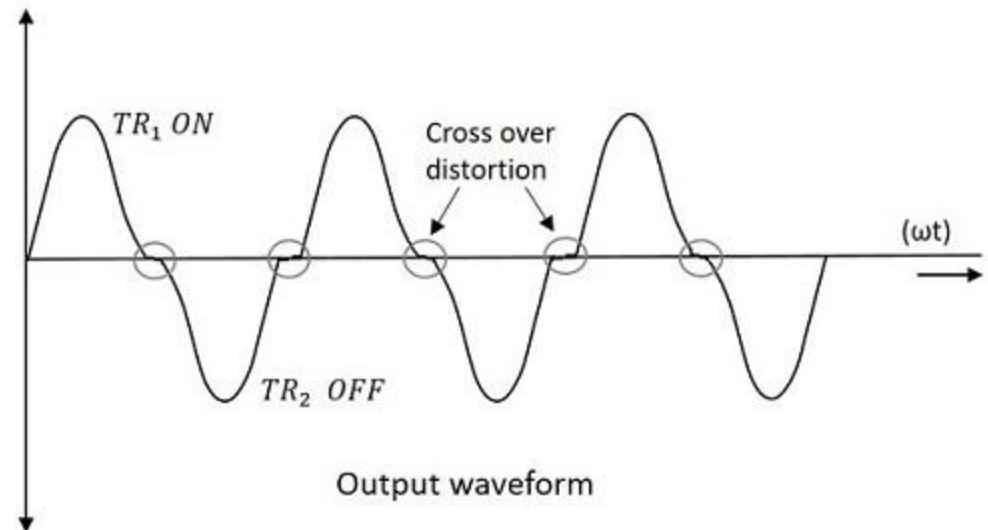
Here, each transistor conducts only half of the cycle ($\theta=180^\circ$)

In fact, each transistor conducts less than half of the cycle ($\theta<180^\circ$) due to the dead-zone

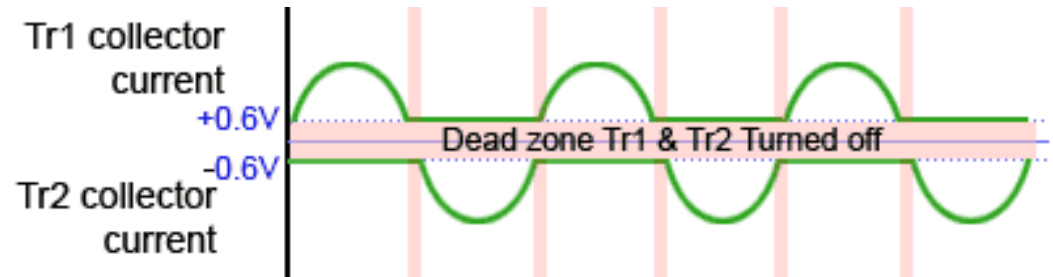
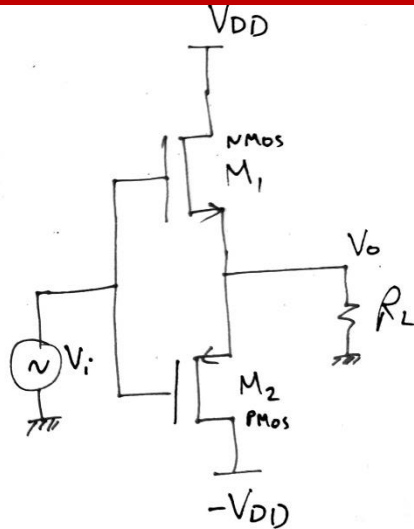
Class B Output Stage: Linearity



- The class B output stage suffers from a dead-zone when $-|V_{THp}| < V_i < V_{THn}$ and where gain ~ 0
- This dead-zone (also called 'cross-over' region) causes some distortion to the signal



Class B Output Stage: Efficiency



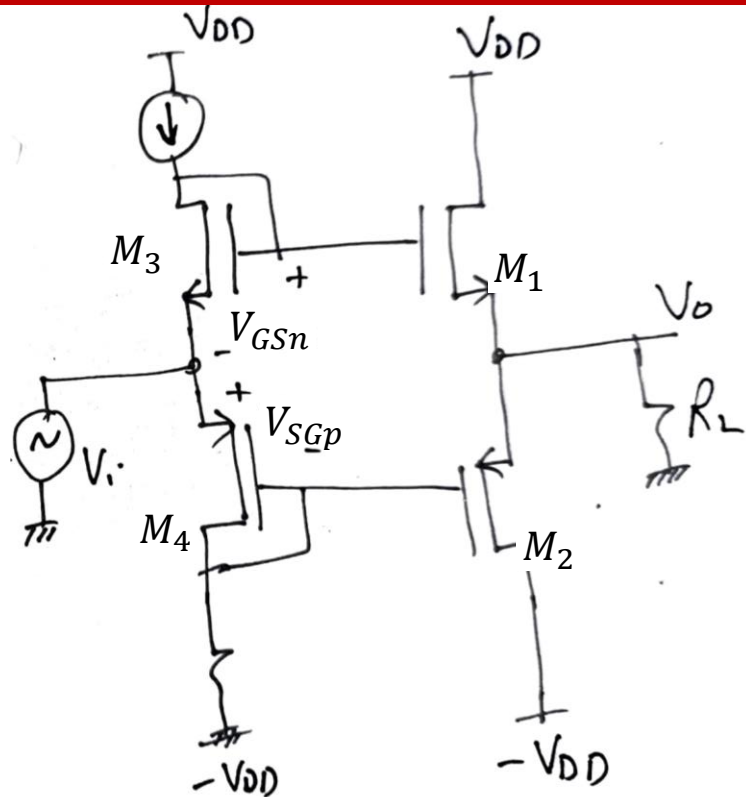
- In class B output stage, each transistor conducts only half of the cycle ($\theta=180^\circ$). In fact, each transistor conducts less than half of the cycle ($\theta<180^\circ$) due to the dead-zone where the gain is ~ 0 .
- Therefore, we expect the efficiency to improve.
- Assuming V_o is a sine wave with amplitude $\hat{V}_o$, and neglecting dead-zone,

Average currents through the transistors: $\overline{I_{M1}} = \overline{I_{M2}} = \frac{\hat{V}_o}{\pi R_L}$

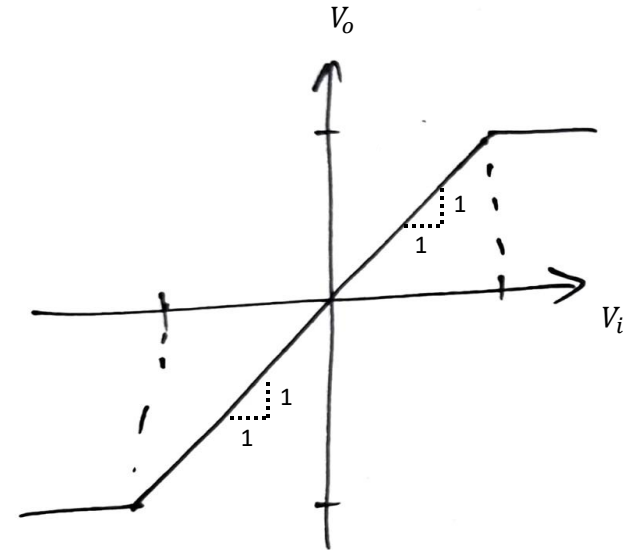
Power Consumption: $V_{DD} \overline{I_{M1}} + V_{DD} \overline{I_{M2}} = \frac{2V_{DD} \hat{V}_o}{\pi R_L}$

Power delivered to load: $\frac{\hat{V}_o^2}{2R_L} \rightarrow \text{Efficiency: } \eta = \frac{\pi}{4} \frac{\hat{V}_o}{V_{DD}} < 78.5\%$

Class AB Output Stage: Push Pull with reduced cross-over



$$V_o \approx V_i$$



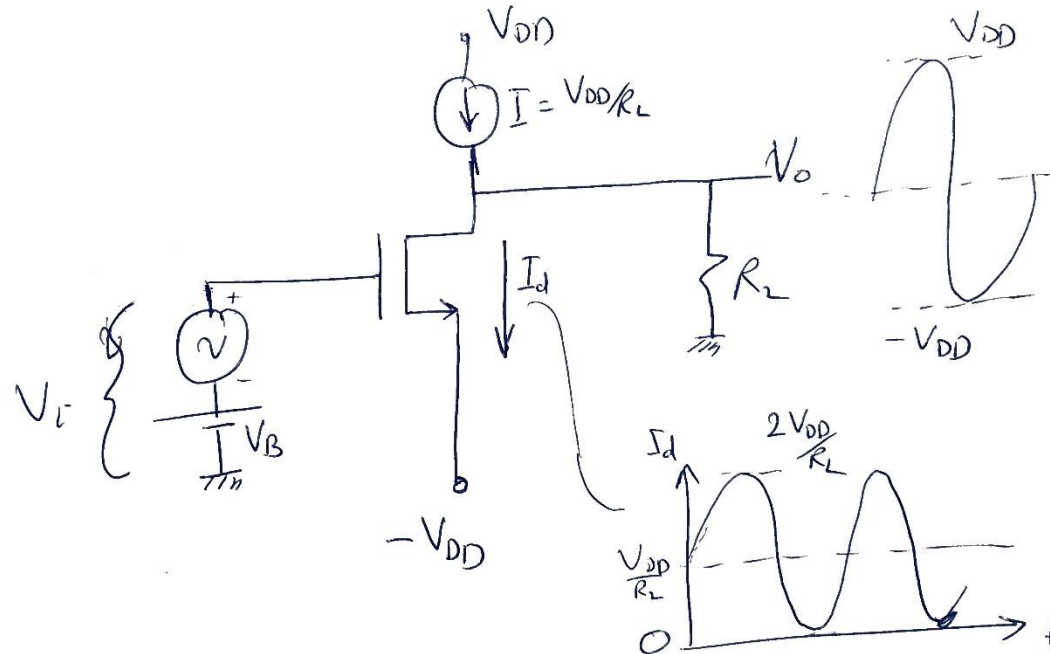
- Class AB attempts to gain the best of class A and class B
- The biasing branch M_3/M_4 shifts the input up and down. Therefore, the input on the gate of M_1 is $V_i + V_{THn}$, and on the gate of M_2 is $V_i - |V_{THp}|$. This eliminates most of the dead-zone.
- Linearity is almost like class A, but its swing is limited by M_3 and M_4 . Calculate O/P swing.
- Efficiency: lower than class B due to lower swing of the output and due to the extra current in M_3/M_4 .

Class C Output Stage: Smaller conduction angle

$$P_{out} = (V_o - \text{rms})^2 / R_L$$

$$P_{in} = (2V_{DD}) * (V_{DD} / R_L)$$

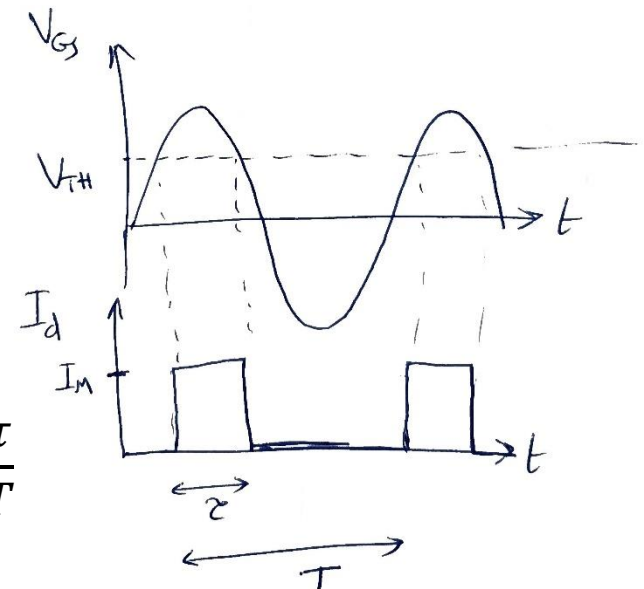
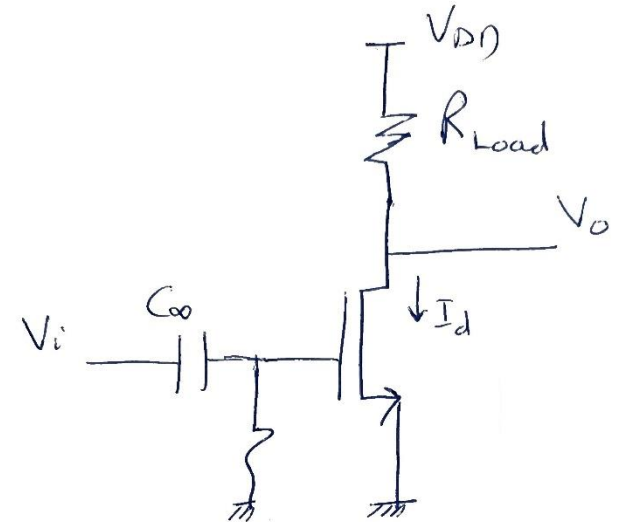
$$\text{Eff} = 25\%$$



- Shown is a Class-A amplifier. Notice that transistor is always conducting current.
- The current flowing in the drain (and thus the current injected in the load) is a sinusoidal current.
- What happens if V_B is set lower, say zero voltage?

Class C Output Stage: Smaller conduction angle

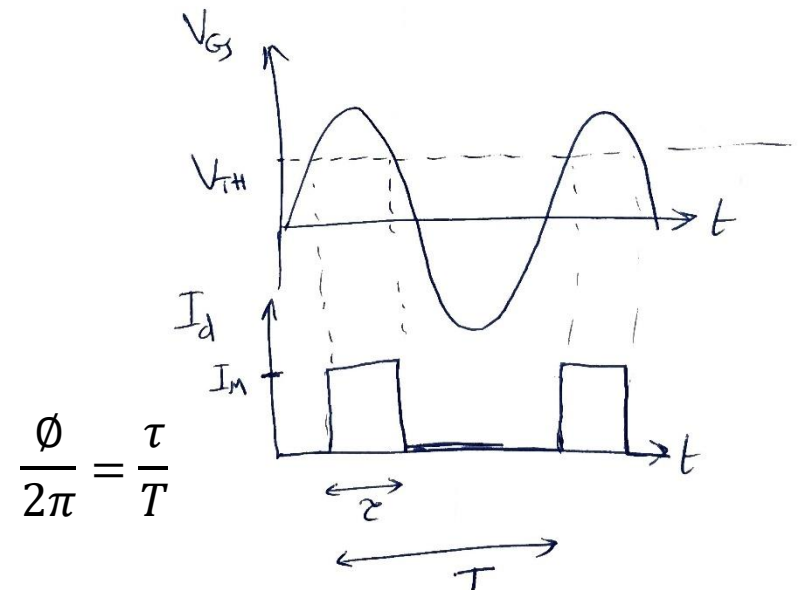
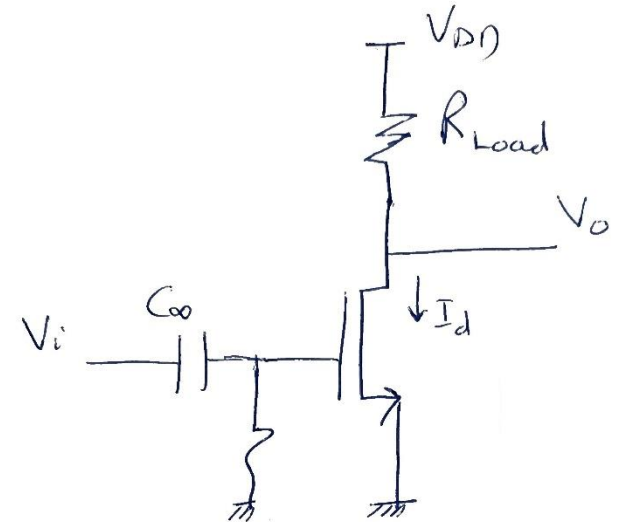
- Assume that the input voltage signal is large enough such that the transistor starts conducting current when V_{in} is close to its peak (i.e. $V_{in-pk} > V_{th}$)
- Thus the transistor will conduct only part of the cycle $\rightarrow$ smaller conduction angle, ϕ .
- Since the transistor conducts only a fraction of the time, efficiency improves. In addition, during conduction, V_{DS} is small $\rightarrow$ less power dissipated in transistor.
- Power drawn from supply: average of $I_d * V_{DD}$
- Power delivered to load: obtain rms value of current. But note that this total power delivered to the load is spread over many frequency harmonics.



$$\frac{\phi}{2\pi} = \frac{\tau}{T}$$

Class C Output Stage: Problem

- Problem: Current in load is pulsed $\rightarrow$ non-linear and full of harmonics.



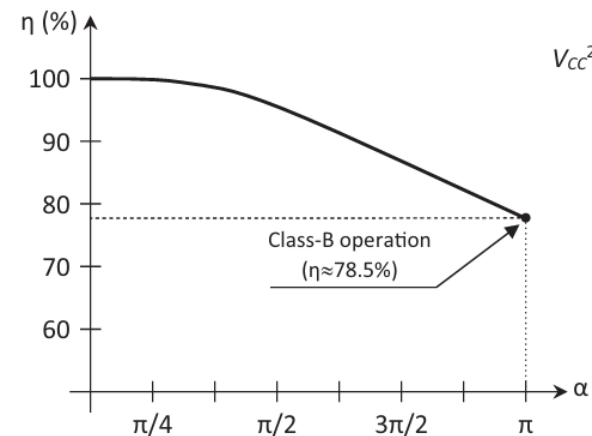
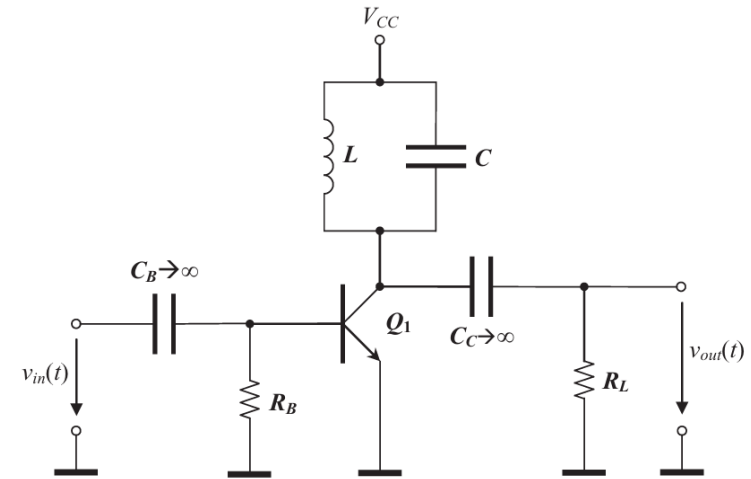
$$\frac{\phi}{2\pi} = \frac{\tau}{T}$$

Class C Output Stage: Tuned Class-C

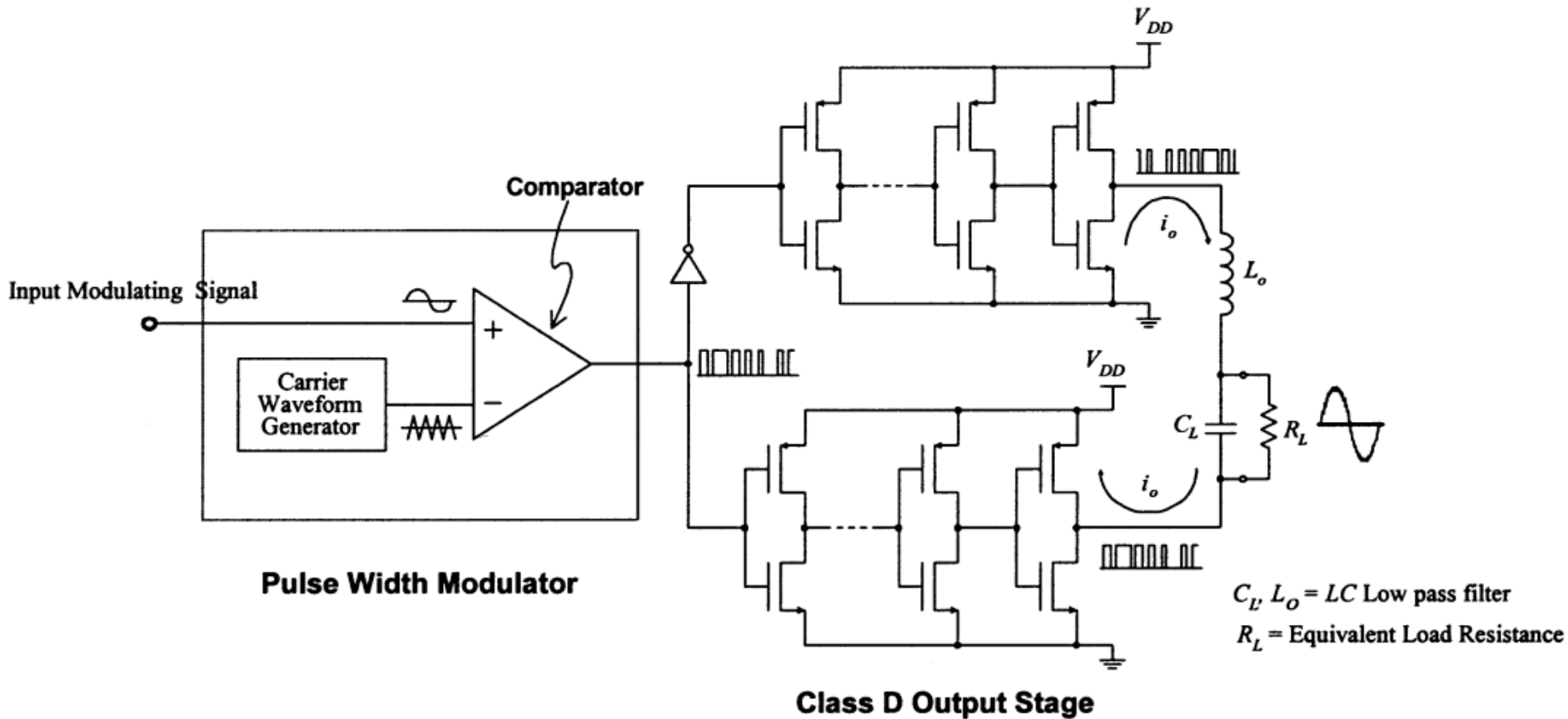
- Assume Q1 is an NFET transistor.
- The LC circuit is tuned to the desired (fundamental) frequency. It attenuates the other harmonics.
- Power delivered to the load: obtain Fourier series of transistor current and take only the fundamental.
- Power drawn from supply:
 - Obtain average value of transistor current.
 - Or, obtain power dissipated in transistor ($I_{ds} * V_{ds} * \tau / T$).

- Equation given by, where α is conduction angle ϕ :

$$\eta = \frac{\frac{\alpha}{2} - \frac{\sin \alpha}{2}}{2 \sin \frac{\alpha}{2} - \alpha \cos \frac{\alpha}{2}}$$



Class D output Stage



- Convert the signal to a Pulse-Width Modulated signal
- Driver has only two levels: $+I_{out}$ and $-I_{out}$.
- Efficiency can be $\sim 90\%$

Class D output Stage

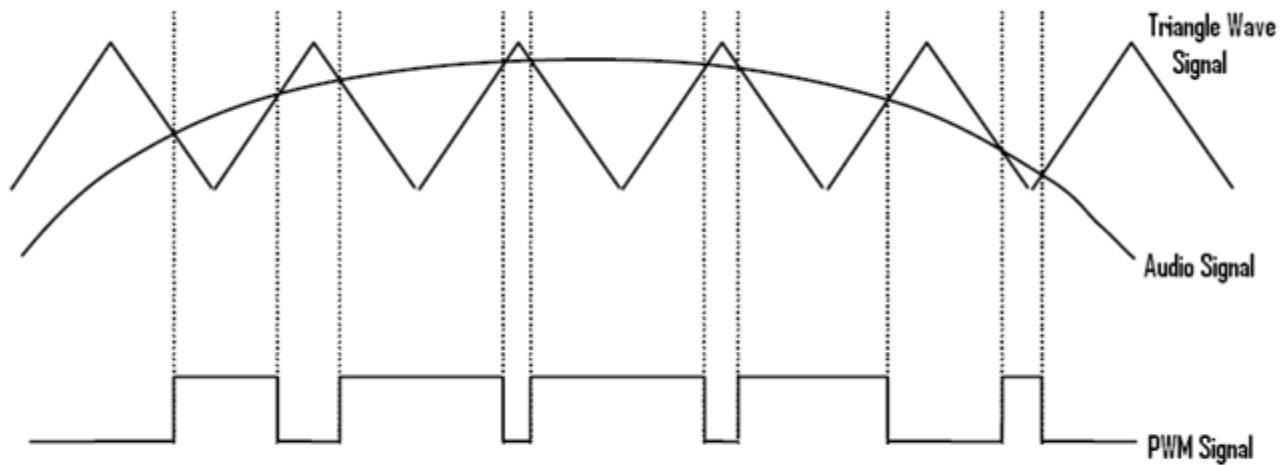


Figure 10: Graphical representation of a PWM signal generation [2]

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